

Dip-Timed Momentum Feed (US) — Investor Simulation

Marketworks Research

\$100,000 deployed 03 Jan 2017 → 30 Dec 2022 · S&P 500 ∪ Nasdaq 100 · max 25 positions Backtest simulation — not live results

\$238k

Final value (net of tax)

+138%

Total return

15.6%

CAGR (net of tax)

11.2%

SPY CAGR

-37.3%

Max drawdown

\$40.0k

Tax paid

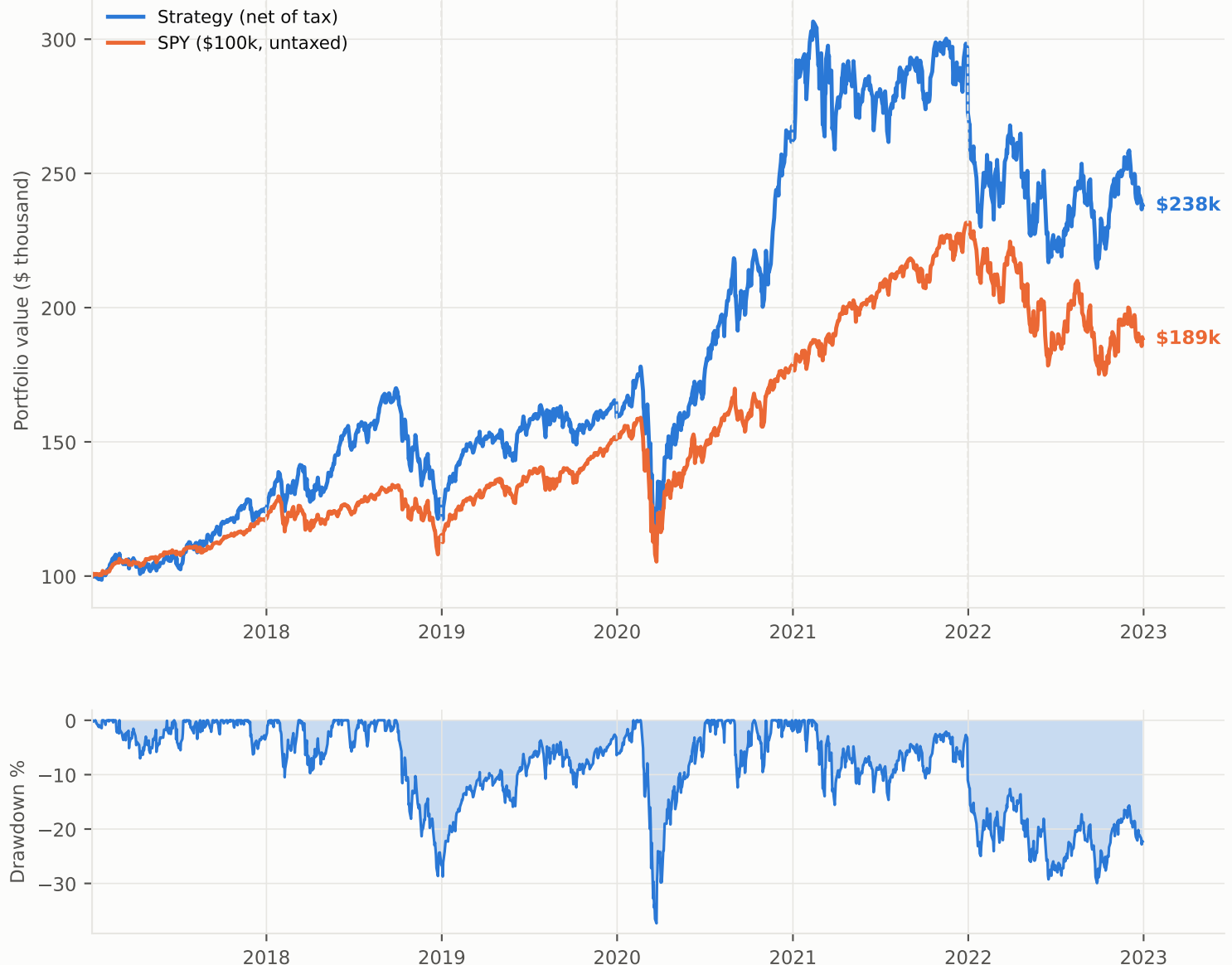
288 / 25

Calls closed / open

49%

Win rate

Growth of \$100,000 (dashed lines: calendar-year ends / tax dates)



Strategy rules

Entry: a stock in the top quartile of the S&P 500 ∪ Nasdaq 100 universe by 126-day risk-adjusted momentum, trading above its 200-day average, falls more than 5% over 5 sessions. Slots (max 25) are filled by momentum rank; each position is 4% of portfolio equity at entry, whole shares. Fills at next-day average price (OHLC/4) with 0.2% slippage each way. Exit: momentum rank decays below the 35th percentile — no trailing stop (a 20% trail subtracts ~4pp CAGR on US data; see RESULTS_US.md). Tax: 25% flat on net realized gains each Dec 31, losses carried forward. No leverage; idle cash earns nothing. Config selected on Indian data, tested here out-of-sample with no parameter retuning.

Returns Detail

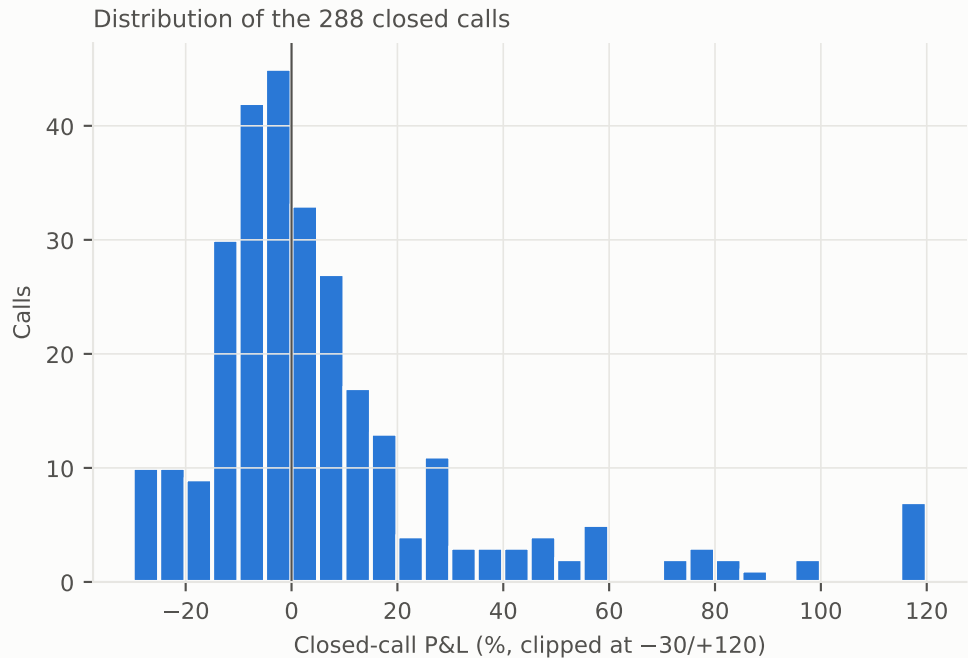
Monthly portfolio returns and calendar-year summary

Monthly returns (%) — net equity, after tax outflows

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2017	+0.4	+5.1	+0.8	-2.6	+1.4	-1.0	+6.4	+4.2	+3.7	+2.6	+2.3	-1.0	+24.2
2018	+9.5	-1.9	-0.9	-0.6	+10.6	+1.9	+1.9	+10.7	+1.8	-14.0	-1.7	-13.1	+0.6
2019	+12.3	+6.1	+1.5	+1.3	-5.7	+7.5	+3.3	+0.9	-4.3	+1.3	+2.9	-1.2	+27.5
2020	+2.6	-6.2	-13.8	+16.3	+6.2	+6.2	+11.7	+12.9	-4.1	-4.3	+21.4	+7.8	+64.8
2021	+5.9	+2.4	-4.6	+5.0	-1.1	-0.4	-0.4	+3.5	-3.8	+7.3	-2.8	-5.8	+4.1
2022	-10.4	+3.0	+3.7	-7.7	+0.5	-8.3	+7.2	+1.8	-9.9	+13.1	+4.6	-7.6	-12.7

Calendar-year summary — 25% flat on net realized gains, losses carried forward

Year end	Realized P&L (\$)	Tax paid (\$)	Loss carried fwd (\$)	Equity after (\$)
2017-12-29	-443	0	443	124,077
2018-12-31	22,944	5,625	0	124,794
2019-12-31	23,545	5,886	0	159,094
2020-12-31	20,773	5,193	0	262,146
2021-12-31	92,983	23,246	0	272,838
2022-12-30	-12,054	0	12,054	238,195



Avg win	+32.5%
Avg loss	-10.4%
Profit factor	2.40
Median hold	90 sessions
Best call	+770% (TSLA)
Worst call	-44% (FCX)
Exit rule	momentum decay (100%)
Sharpe (house conv.)	0.41

Current Holdings — 30 Dec 2022

25 open positions · market value \$237.5k · cash \$648

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Backtest simulation — not live results

Symbol	Entry	Shares	Entry \$	Last \$	Value \$	P&L \$	P&L %	Hold (td)	Mom rank	vs peak
PWR	2021-01-29	161	70.67	141.85	22,838	11,461	+100.7%	484	0.63	-6.9%
AZO	2021-08-30	7	1,557.70	2,466.18	17,263	6,359	+58.3%	337	0.78	-4.4%
PDD	2022-10-11	57	56.59	81.55	4,648	1,423	+44.1%	56	0.60	-10.5%
MPC	2022-06-30	112	76.07	108.07	12,104	3,584	+42.1%	127	0.95	-8.1%
ELV	2021-08-12	22	342.74	485.27	10,676	3,136	+41.6%	349	0.54	-5.9%
IRM	2021-06-29	317	35.13	44.15	13,997	2,862	+25.7%	380	0.44	-10.5%
FSLR	2022-10-21	67	120.72	149.79	10,036	1,947	+24.1%	48	1.00	-13.2%
MCK	2022-06-07	26	314.99	368.95	9,593	1,403	+17.1%	143	0.76	-5.5%
STLD	2022-04-01	130	79.97	92.96	12,085	1,689	+16.2%	188	0.95	-11.2%
AMGN	2022-05-05	14	206.38	235.21	3,293	404	+14.0%	165	0.64	-9.6%
MNST	2022-08-11	213	44.89	50.77	10,813	1,251	+13.1%	98	0.58	-2.4%
NFLX	2022-11-03	50	27.19	29.49	1,474	115	+8.5%	39	0.96	-8.0%
BIIB	2022-10-06	35	262.94	276.92	9,692	489	+5.3%	59	0.71	-9.7%
FICO	2022-11-23	13	583.83	598.58	7,782	192	+2.5%	25	0.87	-4.2%
CAH	2022-11-14	22	71.10	72.52	1,595	31	+2.0%	32	0.99	-4.3%
IBKR	2022-12-14	567	17.62	17.76	10,069	81	+0.8%	11	0.90	-1.9%
ERIE	2022-12-29	40	235.78	233.20	9,328	-103	-1.1%	1	0.91	-1.0%
AXON	2022-12-07	58	170.52	165.93	9,624	-266	-2.7%	16	0.99	-5.0%
SMCI	2022-12-06	1,196	8.44	8.21	9,819	-280	-2.8%	17	1.00	-8.8%
NDAQ	2022-12-12	161	60.91	58.39	9,401	-405	-4.1%	13	0.80	-5.6%
HII	2022-11-07	42	225.12	214.10	8,992	-463	-4.9%	37	0.52	-5.1%
TKO	2022-11-07	140	69.80	64.49	9,028	-744	-7.6%	37	0.64	-15.1%
MRNA	2022-12-13	51	194.74	179.62	9,161	-771	-7.8%	12	0.58	-14.5%
SNPS	2022-08-25	22	361.72	319.29	7,024	-934	-11.7%	88	0.49	-12.1%
EPAM	2022-08-30	22	421.53	327.74	7,210	-2,063	-22.2%	85	0.49	-25.3%

Mom rank = current cross-sectional momentum percentile (exit below 0.35).

vs peak = distance from peak close since entry (informational — no trailing stop in this configuration).

Closed-Call Ledger (1/7)

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All 288 closed calls, most recent first · prices are effective (slippage included)

Backtest simulation — not live results

Symbol	Entry	Exit	Shares	Entry \$	Exit \$	P&L \$	P&L %	Hold	Exit type
ALB	2022-09-23	2022-12-28	35	253.37	202.98	-1,764	-19.9%	66	decay
JKHY	2022-08-18	2022-12-13	51	190.06	175.58	-739	-7.6%	81	decay
VST	2022-07-12	2022-11-07	409	20.99	22.10	451	+5.3%	83	decay
EQT	2022-07-06	2022-11-22	188	30.39	40.80	1,957	+34.2%	98	decay
OXY	2022-06-23	2022-12-12	119	53.50	59.78	747	+11.7%	119	decay
KHC	2022-06-02	2022-10-21	280	29.63	28.91	-204	-2.5%	98	decay
HSY	2022-05-24	2022-11-14	7	191.23	198.58	51	+3.8%	120	decay
COR	2022-05-19	2022-10-11	2	143.77	135.84	-16	-5.5%	99	decay
ATO	2022-05-03	2022-10-06	90	101.90	93.05	-796	-8.7%	108	decay
CF	2022-04-22	2022-12-14	97	90.21	88.13	-202	-2.3%	163	decay
ABBV	2022-04-14	2022-08-25	60	137.80	120.05	-1,065	-12.9%	91	decay
ANET	2022-04-11	2022-06-02	320	32.14	25.86	-2,007	-19.5%	36	decay
FCX	2022-04-07	2022-07-06	225	45.46	25.44	-4,506	-44.0%	60	decay
UNP	2022-04-05	2022-07-11	45	231.48	190.61	-1,839	-17.6%	65	decay
RTX	2022-03-09	2022-08-30	108	88.61	84.23	-473	-4.9%	120	decay
VLO	2022-03-02	2022-12-06	135	73.14	107.97	4,702	+47.6%	193	decay
APA	2022-02-22	2022-12-07	349	27.67	38.37	3,733	+38.6%	200	decay
KEY	2022-01-26	2022-04-05	460	20.11	17.24	-1,322	-14.3%	48	decay
AVGO	2022-01-19	2022-06-30	187	53.03	45.54	-1,400	-14.1%	112	decay
BLDR	2022-01-12	2022-06-23	126	81.70	50.62	-3,917	-38.0%	111	decay
PLD	2022-01-06	2022-05-19	3	137.13	104.63	-97	-23.7%	92	decay
IBKR	2022-01-06	2022-04-07	1	18.82	15.55	-3	-17.4%	63	decay
FDS	2022-01-06	2022-05-03	24	428.45	379.19	-1,182	-11.5%	80	decay
AMD	2022-01-05	2022-04-14	64	139.78	95.49	-2,835	-31.7%	69	decay
DDOG	2021-12-17	2022-04-21	68	165.40	128.27	-2,525	-22.4%	85	decay
MRVL	2021-12-17	2022-06-07	139	81.95	58.89	-3,206	-28.1%	117	decay
NVDA	2021-12-14	2022-05-05	154	27.96	19.19	-1,351	-31.4%	98	decay
MPWR	2021-12-13	2022-01-26	24	479.11	368.85	-2,646	-23.0%	30	decay
TSLA	2021-11-11	2022-05-24	6	361.22	212.59	-892	-41.1%	133	decay
GNRC	2021-11-04	2021-12-13	23	455.60	346.69	-2,505	-23.9%	26	decay
DASH	2021-11-03	2021-12-13	49	203.07	154.53	-2,378	-23.9%	27	decay
BRO	2021-11-01	2022-10-26	25	60.53	53.73	-170	-11.2%	248	decay
PANW	2021-10-27	2022-11-07	145	82.31	71.50	-1,568	-13.1%	259	decay
FTNT	2021-09-14	2022-08-12	169	60.15	53.53	-1,118	-11.0%	230	decay
ALNY	2021-09-13	2021-12-17	61	187.85	188.19	21	+0.2%	68	decay
INVH	2021-09-13	2022-03-02	335	34.63	32.88	-586	-5.1%	118	decay
REGN	2021-09-10	2022-01-06	18	647.54	593.75	-968	-8.3%	82	decay
STE	2021-08-25	2022-08-11	55	205.71	203.43	-125	-1.1%	242	decay
COF	2021-08-23	2021-11-03	71	156.29	140.07	-1,152	-10.4%	51	decay
STLD	2021-08-20	2021-11-11	38	62.47	61.62	-32	-1.4%	58	decay
ALB	2021-08-19	2022-02-22	55	201.12	181.43	-1,083	-9.8%	128	decay
AXP	2021-08-19	2021-09-07	74	150.30	149.75	-41	-0.4%	12	decay
MRNA	2021-08-12	2021-12-28	29	394.50	244.25	-4,357	-38.1%	95	decay
HCA	2021-08-11	2022-01-19	49	230.63	237.97	360	+3.2%	111	decay
TT	2021-08-11	2021-11-04	62	182.07	169.80	-761	-6.7%	60	decay
GNRC	2021-07-30	2021-08-18	26	420.70	390.15	-794	-7.3%	13	decay
FIX	2021-07-19	2021-09-02	149	70.89	74.21	495	+4.7%	33	decay

Closed-Call Ledger (2/7)

All 288 closed calls, most recent first · prices are effective (slippage included)

Backtest simulation — not live results

Symbol	Entry	Exit	Shares	Entry \$	Exit \$	P&L \$	P&L %	Hold	Exit type
NUE	2021-07-19	2022-09-23	127	83.33	99.35	2,034	+19.2%	299	decay
DELL	2021-07-16	2022-04-01	252	43.02	43.97	238	+2.2%	180	decay
APO	2021-07-15	2022-04-08	203	53.94	52.96	-200	-1.8%	186	decay
FANG	2021-07-13	2021-08-05	154	72.79	63.42	-1,444	-12.9%	17	decay
GS	2021-07-09	2021-12-30	33	327.30	345.69	607	+5.6%	121	decay
ADM	2021-06-15	2021-08-23	198	56.47	51.64	-955	-8.5%	48	decay
F	2021-06-11	2021-08-18	990	11.36	9.63	-1,715	-15.2%	47	decay
DVN	2021-06-11	2021-09-02	490	22.93	22.69	-115	-1.0%	58	decay
RF	2021-06-10	2021-07-15	637	17.78	15.88	-1,208	-10.7%	24	decay
MOS	2021-06-10	2021-07-15	364	31.11	27.25	-1,406	-12.4%	24	decay
DE	2021-05-24	2021-08-23	32	336.83	334.66	-70	-0.7%	63	decay
RL	2021-05-24	2021-06-15	101	108.45	108.24	-21	-0.2%	15	decay
WSM	2021-05-19	2021-07-28	134	74.72	70.61	-550	-5.5%	48	decay
HPQ	2021-05-13	2021-08-10	129	26.67	24.55	-274	-8.0%	61	decay
OXY	2021-05-12	2021-07-19	477	23.58	22.89	-329	-2.9%	46	decay
ASML	2021-05-05	2022-01-12	18	615.42	708.76	1,680	+15.2%	175	decay
TECH	2021-05-05	2022-01-05	110	102.72	110.20	823	+7.3%	170	decay
STX	2021-05-05	2021-10-18	148	75.96	70.56	-800	-7.1%	115	decay
BKR	2021-04-13	2021-06-11	637	17.58	22.84	3,351	+29.9%	42	decay
TPR	2021-03-30	2021-08-12	293	35.36	39.03	1,076	+10.4%	94	decay
NWSA	2021-03-30	2021-08-11	394	24.32	23.03	-508	-5.3%	93	decay
TPL	2021-03-25	2021-09-01	60	155.71	141.12	-875	-9.4%	111	decay
ZBRA	2021-03-09	2021-10-07	6	471.02	507.47	219	+7.7%	148	decay
ALGN	2021-03-08	2021-04-30	20	516.28	599.61	1,667	+16.1%	38	decay
AMAT	2021-03-04	2021-08-13	37	107.22	123.92	618	+15.6%	113	decay
ON	2021-02-26	2021-06-11	49	39.37	37.84	-75	-3.9%	73	decay
MRNA	2021-02-23	2021-06-03	79	147.90	192.80	3,547	+30.4%	70	decay
AES	2021-02-23	2021-06-25	52	21.91	21.43	-25	-2.2%	86	decay
KEYS	2021-02-22	2021-07-13	45	140.25	157.36	770	+12.2%	98	decay
COHR	2021-02-19	2021-05-03	131	90.50	67.34	-3,034	-25.6%	50	decay
DVA	2021-02-08	2021-05-19	83	110.04	121.25	930	+10.2%	70	decay
AXON	2021-02-03	2021-05-04	56	167.41	144.84	-1,264	-13.5%	62	decay
CPRT	2021-01-19	2021-02-26	72	29.10	27.05	-148	-7.1%	27	decay
MRNA	2020-12-22	2021-01-19	59	132.58	123.25	-551	-7.0%	17	decay
MSTR	2020-12-10	2021-06-03	13	28.80	49.30	266	+71.2%	119	decay
BG	2020-12-03	2021-07-16	177	52.28	65.28	2,301	+24.9%	154	decay
MTD	2020-11-25	2021-03-08	6	1,127.67	1,048.50	-475	-7.0%	68	decay
AWK	2020-11-24	2021-02-23	41	138.75	136.43	-95	-1.7%	60	decay
GNRC	2020-11-19	2021-05-11	41	215.66	298.73	3,406	+38.5%	117	decay
TMO	2020-11-19	2021-02-23	16	449.30	450.75	23	+0.3%	63	decay
FSLR	2020-11-11	2021-02-22	72	83.60	87.67	293	+4.9%	68	decay
POOL	2020-11-10	2021-02-03	27	309.09	331.83	614	+7.4%	57	decay
XYZ	2020-10-22	2021-03-30	48	177.86	208.20	1,456	+17.1%	108	decay
URI	2020-10-15	2021-08-23	48	181.16	323.03	6,810	+78.3%	214	decay
PSKY	2020-10-15	2021-04-13	358	24.18	35.70	4,125	+47.6%	122	decay
FCX	2020-09-25	2021-07-09	450	14.10	33.67	8,807	+138.8%	197	decay
META	2020-09-21	2020-11-18	25	245.78	271.00	631	+10.3%	42	decay

Closed-Call Ledger (3/7)

Marketworks Research

All 288 closed calls, most recent first · prices are effective (slippage included)

Backtest simulation — not live results

Symbol	Entry	Exit	Shares	Entry \$	Exit \$	P&L \$	P&L %	Hold	Exit type
MELI	2020-09-17	2021-03-30	7	980.92	1,413.49	3,028	+44.1%	133	decay
LULU	2020-09-15	2020-12-10	1	317.01	365.90	49	+15.4%	61	decay
TSCO	2020-09-14	2020-11-25	286	25.25	24.28	-278	-3.9%	52	decay
AMZN	2020-09-11	2020-12-22	49	158.14	159.82	82	+1.1%	71	decay
CMG	2020-09-11	2021-03-04	146	26.16	27.37	177	+4.6%	119	decay
ZS	2020-09-10	2021-03-08	42	137.69	172.70	1,471	+25.4%	122	decay
SHOP	2020-09-08	2020-11-10	88	92.65	89.74	-256	-3.1%	45	decay
CVNA	2020-09-08	2021-02-08	158	36.19	57.62	3,387	+59.2%	105	decay
EBAY	2020-09-03	2020-11-10	131	49.09	42.38	-878	-13.7%	47	decay
PDD	2020-08-25	2021-04-30	33	85.69	135.30	1,637	+57.9%	171	decay
CRWD	2020-08-10	2021-03-25	202	25.01	44.37	3,911	+77.4%	157	decay
AOS	2020-08-05	2020-12-03	182	43.23	49.59	1,158	+14.7%	84	decay
MSCI	2020-08-04	2020-10-06	20	345.26	329.67	-312	-4.5%	44	decay
DLR	2020-06-05	2020-09-11	57	113.37	119.11	327	+5.1%	68	decay
SJM	2020-05-21	2020-09-15	4	89.07	92.96	16	+4.4%	80	decay
SBAC	2020-05-05	2020-09-11	17	266.78	279.46	216	+4.8%	90	decay
GEN	2020-03-18	2020-09-21	307	14.12	18.20	1,252	+28.9%	129	decay
AAPL	2020-02-25	2021-02-19	94	71.22	126.16	5,164	+77.1%	249	decay
LRCX	2020-02-24	2020-09-10	206	27.95	28.18	47	+0.8%	139	decay
ALLE	2020-02-21	2020-05-05	53	118.33	88.49	-1,582	-25.2%	51	decay
SMCI	2020-02-19	2020-09-08	2,382	2.76	2.52	-555	-8.5%	140	decay
INSM	2020-02-12	2020-08-05	222	31.20	26.83	-970	-14.0%	121	decay
APO	2020-02-03	2020-09-25	165	39.46	38.27	-196	-3.0%	164	decay
AMD	2020-01-31	2021-01-29	137	47.79	86.46	5,299	+80.9%	251	decay
AMGN	2020-01-27	2020-09-17	36	183.25	205.42	798	+12.1%	163	decay
CVS	2020-01-27	2020-05-21	5	57.17	51.60	-28	-9.7%	81	decay
ABBV	2020-01-27	2020-09-14	101	65.56	71.48	598	+9.0%	160	decay
LITE	2020-01-16	2020-09-03	83	77.51	77.38	-10	-0.2%	160	decay
TGT	2020-01-16	2021-12-17	66	97.79	193.41	6,311	+97.8%	485	decay
CVNA	2020-01-06	2020-02-11	346	18.43	17.47	-331	-5.2%	25	decay
PDD	2019-11-26	2020-02-24	173	36.52	33.19	-576	-9.1%	59	decay
TSLA	2019-11-25	2021-05-24	278	22.71	197.62	48,625	+770.3%	375	decay
TER	2019-11-22	2021-05-04	9	60.54	119.34	529	+97.1%	362	decay
SWKS	2019-11-21	2020-11-18	77	81.52	119.71	2,941	+46.9%	250	decay
EQR	2019-11-08	2020-02-20	95	64.82	66.44	154	+2.5%	69	decay
ES	2019-11-08	2020-08-25	42	61.98	68.03	254	+9.8%	199	decay
NEE	2019-11-08	2020-09-08	131	47.18	59.89	1,665	+26.9%	208	decay
NEM	2019-11-08	2020-10-09	204	30.36	52.75	4,566	+73.7%	231	decay
O	2019-11-08	2019-12-13	115	53.68	50.88	-322	-5.2%	24	decay
DOC	2019-11-07	2019-12-13	260	23.84	22.59	-323	-5.2%	25	decay
MLM	2019-11-06	2020-02-12	25	242.07	248.07	150	+2.5%	66	decay
PLD	2019-10-29	2020-11-24	68	71.82	84.62	870	+17.8%	271	decay
HSY	2019-10-28	2020-01-16	47	121.72	126.95	246	+4.3%	55	decay
FISV	2019-10-23	2020-06-05	60	101.06	108.06	420	+6.9%	155	decay
LHX	2019-10-21	2019-11-26	34	175.79	174.48	-45	-0.8%	26	decay
PODD	2019-10-07	2020-10-22	39	157.25	243.39	3,360	+54.8%	264	decay
BX	2019-09-30	2020-08-05	157	39.07	42.90	600	+9.8%	214	decay

Closed-Call Ledger (4/7)

Marketworks Research

All 288 closed calls, most recent first · prices are effective (slippage included)

Backtest simulation — not live results

Symbol	Entry	Exit	Shares	Entry \$	Exit \$	P&L \$	P&L %	Hold	Exit type
BALL	2019-09-10	2019-11-08	93	67.40	59.91	-697	-11.1%	43	decay
TSN	2019-09-09	2019-11-11	93	69.80	67.50	-214	-3.3%	45	decay
CSGP	2019-08-05	2019-12-13	106	58.85	57.35	-160	-2.6%	92	decay
CDNS	2019-08-05	2019-10-23	91	68.53	63.04	-500	-8.0%	56	decay
BR	2019-08-01	2019-11-07	43	113.30	105.40	-339	-7.0%	69	decay
MSCI	2019-07-26	2019-10-23	29	219.31	201.81	-507	-8.0%	62	decay
FDS	2019-07-22	2019-09-27	21	261.59	224.80	-772	-14.1%	48	decay
IR	2019-07-10	2019-10-29	153	31.70	31.51	-28	-0.6%	78	decay
ARES	2019-06-26	2022-03-09	299	20.50	60.54	11,974	+195.4%	681	decay
SHOP	2019-06-26	2019-11-25	212	28.94	31.52	547	+8.9%	106	decay
AXON	2019-06-18	2019-10-07	89	68.22	54.44	-1,227	-20.2%	77	decay
QCOM	2019-05-22	2019-06-21	100	58.91	61.85	293	+5.0%	21	decay
CVNA	2019-05-17	2019-11-07	439	13.60	14.78	518	+8.7%	121	decay
TDG	2019-05-16	2020-03-18	17	342.03	208.55	-2,269	-39.0%	211	decay
FAST	2019-05-13	2019-07-16	443	13.39	12.93	-205	-3.5%	44	decay
MRVL	2019-05-10	2019-10-17	261	22.61	23.12	132	+2.2%	111	decay
SMCI	2019-05-09	2019-06-13	2,716	2.12	1.77	-959	-16.7%	24	decay
GRMN	2019-05-02	2019-07-31	66	67.49	69.09	105	+2.4%	62	decay
KEYS	2019-05-01	2020-01-31	70	87.20	93.86	466	+7.6%	190	decay
ESS	2019-04-23	2019-12-13	24	215.34	235.83	492	+9.5%	164	decay
WAT	2019-04-18	2019-05-22	25	239.70	206.08	-840	-14.0%	23	decay
DOC	2019-04-18	2019-07-25	298	20.06	21.83	527	+8.8%	67	decay
VEEV	2019-04-18	2019-11-01	46	127.88	143.45	716	+12.2%	137	decay
O	2019-04-18	2019-06-26	128	46.64	47.52	113	+1.9%	47	decay
DHR	2019-04-18	2019-11-06	54	109.50	115.13	304	+5.1%	140	decay
WELL	2019-04-18	2019-11-22	8	58.61	69.13	84	+17.9%	152	decay
ZS	2019-04-05	2019-09-09	95	63.59	62.94	-61	-1.0%	107	decay
KLAC	2019-03-29	2020-02-25	529	10.88	14.10	1,705	+29.6%	228	decay
INSM	2019-03-20	2019-07-10	208	28.88	23.23	-1,177	-19.6%	77	decay
ERIE	2019-03-13	2019-10-25	1	151.58	160.68	9	+6.0%	158	decay
BA	2019-03-12	2019-04-09	5	371.38	360.45	-55	-2.9%	20	decay
VST	2019-03-11	2019-05-09	271	21.54	21.25	-78	-1.3%	42	decay
AON	2019-03-06	2020-08-04	37	156.40	191.08	1,283	+22.2%	357	decay
ELV	2019-03-05	2019-04-12	22	267.50	238.27	-643	-10.9%	28	decay
WDAY	2019-03-05	2019-08-05	32	181.33	188.50	229	+4.0%	106	decay
EW	2019-02-27	2020-02-03	105	56.15	73.41	1,813	+30.8%	235	decay
OMC	2019-02-12	2019-04-26	101	56.82	62.07	530	+9.2%	51	decay
VZ	2019-01-30	2019-04-26	110	35.04	37.22	240	+6.2%	60	decay
LLY	2018-12-18	2019-04-23	50	97.95	106.18	412	+8.4%	85	decay
KDP	2018-11-21	2019-05-17	247	21.63	24.20	636	+11.9%	120	decay
ORLY	2018-11-21	2019-05-15	234	22.83	23.52	162	+3.0%	118	decay
CIEN	2018-11-15	2019-05-10	170	33.53	34.80	214	+3.8%	119	decay
J	2018-11-14	2019-02-08	99	57.71	50.47	-717	-12.6%	57	decay
MO	2018-11-13	2018-12-28	3	33.58	27.57	-18	-17.9%	30	decay
JKHY	2018-11-12	2019-02-07	40	128.86	119.95	-356	-6.9%	58	decay
TECH	2018-11-05	2019-04-17	128	41.09	46.29	666	+12.7%	111	decay
FTNT	2018-10-31	2019-04-02	348	16.38	17.02	226	+4.0%	103	decay

Closed-Call Ledger (5/7)

All 288 closed calls, most recent first · prices are effective (slippage included)

Backtest simulation — not live results

Symbol	Entry	Exit	Shares	Entry \$	Exit \$	P&L \$	P&L %	Hold	Exit type
WST	2018-10-31	2019-02-04	36	103.96	101.43	-91	-2.4%	63	decay
TTD	2018-10-30	2019-09-10	398	11.09	21.02	3,949	+89.4%	215	decay
AZO	2018-10-22	2020-01-07	8	733.19	1,151.58	3,347	+57.1%	303	decay
LOW	2018-10-22	2018-11-21	45	85.58	75.60	-449	-11.7%	22	decay
AMD	2018-10-11	2019-03-08	218	25.25	21.58	-800	-14.5%	100	decay
FIX	2018-09-20	2019-01-30	126	52.22	45.52	-845	-12.8%	89	decay
WAB	2018-09-18	2018-11-01	65	100.71	79.76	-1,362	-20.8%	32	decay
AES	2018-09-05	2019-05-13	668	10.05	12.38	1,555	+23.2%	171	decay
CMG	2018-08-30	2019-11-06	682	9.73	14.86	3,497	+52.7%	298	decay
PSX	2018-08-20	2018-10-22	73	85.15	75.51	-704	-11.3%	44	decay
AXON	2018-08-01	2018-11-08	87	68.60	51.56	-1,482	-24.8%	70	decay
DPZ	2018-07-25	2018-12-18	20	239.37	221.01	-367	-7.7%	101	decay
DXCM	2018-06-27	2019-03-27	195	23.48	29.43	1,161	+25.4%	187	decay
PTC	2018-06-26	2018-10-31	57	93.49	81.30	-695	-13.0%	89	decay
COP	2018-06-18	2018-11-09	15	50.99	51.05	1	+0.1%	102	decay
WBD	2018-05-10	2019-03-18	217	23.10	27.34	920	+18.4%	213	decay
MPC	2018-05-04	2018-10-22	73	59.82	57.82	-146	-3.3%	118	decay
CVNA	2018-04-25	2019-01-07	555	5.73	6.49	421	+13.2%	176	decay
STX	2018-04-25	2018-08-30	121	42.80	40.23	-311	-6.0%	89	decay
NFLX	2018-04-03	2018-10-31	163	28.52	30.09	257	+5.5%	148	decay
AMZN	2018-03-28	2019-02-04	72	71.65	81.33	697	+13.5%	214	decay
DECK	2018-03-15	2019-08-05	314	15.16	24.08	2,801	+58.9%	349	decay
DE	2018-03-06	2018-06-18	6	140.22	130.45	-59	-7.0%	72	decay
CFG	2018-03-01	2018-06-27	161	31.12	28.46	-428	-8.6%	82	decay
HLT	2018-02-28	2018-07-17	61	78.60	78.38	-13	-0.3%	96	decay
ISRG	2018-02-02	2019-04-17	37	140.48	181.09	1,502	+28.9%	302	decay
DHI	2018-01-30	2018-05-10	125	43.82	40.11	-465	-8.5%	70	decay
TXN	2018-01-25	2018-09-10	61	87.41	87.09	-20	-0.4%	157	decay
INTC	2018-01-10	2018-08-21	87	36.14	40.28	360	+11.5%	154	decay
TKO	2017-12-26	2019-04-01	136	26.71	78.66	7,065	+194.5%	316	decay
RL	2017-12-18	2018-11-21	59	82.68	97.90	898	+18.4%	234	decay
PYPL	2017-12-05	2018-10-11	1	70.37	75.48	5	+7.3%	214	decay
ALGN	2017-12-05	2018-10-30	20	228.08	219.67	-168	-3.7%	227	decay
COHR	2017-12-01	2018-01-24	104	46.41	46.69	30	+0.6%	35	decay
XYZ	2017-11-28	2019-03-05	85	41.58	75.15	2,853	+80.7%	316	decay
HII	2017-11-16	2018-05-04	25	198.62	174.82	-595	-12.0%	115	decay
IQV	2017-11-14	2018-02-23	48	103.04	99.00	-194	-3.9%	68	decay
GNRC	2017-11-08	2018-03-15	101	49.30	46.29	-304	-6.1%	86	decay
VST	2017-11-02	2017-12-14	261	16.31	14.73	-413	-9.7%	29	decay
GM	2017-11-01	2018-03-06	28	37.16	33.08	-114	-11.0%	84	decay
AON	2017-10-31	2017-12-11	36	133.65	128.38	-190	-4.0%	28	decay
ABBV	2017-10-27	2018-04-23	74	64.38	65.65	94	+2.0%	120	decay
PODD	2017-10-27	2018-11-14	82	58.54	80.13	1,771	+36.9%	264	decay
TPL	2017-10-24	2018-11-14	123	39.18	55.94	2,061	+42.8%	267	decay
AMT	2017-09-26	2017-12-27	28	108.18	112.58	123	+4.1%	64	decay
SBAC	2017-09-26	2018-03-01	35	129.76	141.97	427	+9.4%	107	decay
CNC	2017-09-21	2019-02-07	104	45.38	63.92	1,928	+40.8%	346	decay

Closed-Call Ledger (6/7)

Marketworks Research

All 288 closed calls, most recent first · prices are effective (slippage included)

Backtest simulation — not live results

Symbol	Entry	Exit	Shares	Entry \$	Exit \$	P&L \$	P&L %	Hold	Exit type
CCEP	2017-09-15	2017-11-02	137	32.35	31.14	-165	-3.7%	34	decay
TTD	2017-08-11	2017-11-10	17	5.44	5.12	-5	-5.9%	64	decay
IT	2017-08-10	2017-11-06	28	118.61	118.29	-9	-0.3%	61	decay
EXPE	2017-08-07	2017-10-30	30	144.55	117.34	-816	-18.8%	59	decay
ALNY	2017-06-30	2018-04-03	46	79.34	100.87	990	+27.1%	189	decay
APO	2017-06-27	2018-03-27	225	18.84	23.09	955	+22.5%	188	decay
LRCX	2017-06-15	2018-04-20	203	13.24	17.10	785	+29.2%	213	decay
KLAC	2017-06-14	2017-08-03	489	8.73	7.77	-471	-11.0%	35	decay
AAPL	2017-06-13	2018-02-02	124	34.08	38.11	500	+11.8%	162	decay
AMAT	2017-06-13	2017-12-05	103	40.80	45.38	472	+11.2%	122	decay
IDXX	2017-06-13	2017-09-21	26	160.45	156.63	-99	-2.4%	70	decay
EA	2017-06-13	2017-11-08	39	106.81	110.00	125	+3.0%	104	decay
MAR	2017-06-13	2018-06-26	44	95.75	121.08	1,115	+26.5%	261	decay
DLR	2017-06-12	2017-10-27	51	82.88	85.76	147	+3.5%	97	decay
VEEV	2017-06-12	2017-09-26	71	59.68	54.69	-354	-8.3%	74	decay
VRTX	2017-06-12	2017-12-01	34	122.58	141.83	654	+15.7%	121	decay
MELI	2017-05-22	2017-10-13	14	276.41	245.26	-436	-11.3%	101	decay
GEN	2017-05-18	2017-08-07	282	14.45	14.00	-128	-3.1%	55	decay
TSLA	2017-05-18	2017-10-10	197	20.70	23.35	522	+12.8%	100	decay
NCLH	2017-05-15	2017-09-21	80	51.65	54.32	214	+5.2%	90	decay
BLDR	2017-05-10	2017-09-15	273	15.11	16.26	314	+7.6%	89	decay
INSM	2017-05-09	2017-05-30	239	17.20	15.62	-378	-9.2%	14	decay
CHTR	2017-05-04	2017-10-12	12	324.93	357.84	395	+10.1%	112	decay
NRG	2017-04-21	2018-01-22	293	14.05	22.15	2,374	+57.7%	189	decay
ZBRA	2017-04-17	2017-11-09	45	88.45	104.83	737	+18.5%	145	decay
SHOP	2017-04-06	2017-11-06	604	6.84	9.97	1,889	+45.7%	148	decay
MTB	2017-03-22	2017-06-06	29	118.00	119.33	39	+1.1%	52	decay
ON	2017-03-01	2017-06-30	258	15.35	14.12	-317	-8.0%	85	decay
COHR	2017-02-24	2017-05-03	114	37.02	28.11	-1,016	-24.1%	47	decay
URI	2017-02-24	2017-05-18	29	117.72	101.25	-477	-14.0%	58	decay
CF	2017-02-22	2017-04-21	163	26.44	20.57	-957	-22.2%	41	decay
TPL	2017-02-17	2017-04-11	143	29.51	27.94	-224	-5.3%	36	decay
MU	2017-02-15	2018-08-30	188	22.51	50.95	5,347	+126.4%	388	decay
MET	2017-02-06	2017-05-02	121	33.75	33.84	12	+0.3%	59	decay
SCHW	2017-02-03	2017-05-18	113	35.53	33.96	-177	-4.4%	72	decay
HBAN	2017-02-03	2017-05-18	446	9.05	8.33	-322	-8.0%	72	decay
DAL	2017-02-02	2017-05-01	96	42.02	40.74	-123	-3.0%	60	decay
PWR	2017-02-02	2017-05-15	115	35.00	31.88	-360	-8.9%	70	decay
MSI	2017-02-02	2017-02-28	57	70.12	68.73	-79	-2.0%	17	decay
RF	2017-01-24	2017-05-18	396	9.96	9.88	-32	-0.8%	80	decay
STT	2017-01-24	2017-05-16	65	60.57	62.85	148	+3.8%	78	decay
GS	2017-01-20	2017-05-18	20	190.46	176.47	-280	-7.3%	82	decay
C	2017-01-20	2017-06-05	93	42.30	45.89	334	+8.5%	93	decay
ANET	2017-01-20	2018-08-01	700	5.65	16.20	7,383	+186.8%	385	decay
NTRS	2017-01-19	2017-05-18	61	64.75	66.70	119	+3.0%	83	decay
STLD	2017-01-13	2017-05-15	133	29.76	28.60	-154	-3.9%	83	decay
EME	2017-01-12	2017-03-22	60	65.48	58.44	-422	-10.8%	47	decay

Closed-Call Ledger (7/7)

All 288 closed calls, most recent first · prices are effective (slippage included)

Marketworks Research
Backtest simulation — not live results

Symbol	Entry	Exit	Shares	Entry \$	Exit \$	P&L \$	P&L %	Hold	Exit type
STX	2017-01-11	2017-05-19	157	25.42	29.34	615	+15.4%	89	decay
MPC	2017-01-11	2017-06-21	107	37.23	39.63	257	+6.5%	111	decay
VLO	2017-01-11	2017-05-01	86	46.26	44.93	-115	-2.9%	75	decay
AMD	2017-01-05	2017-05-24	350	11.42	10.91	-180	-4.5%	96	decay
NVDA	2017-01-04	2018-08-20	1,564	2.56	6.06	5,486	+137.2%	409	decay
LITE	2017-01-04	2017-04-06	106	37.69	48.48	1,144	+28.6%	64	decay

This is a research backtest, not a live track record.

Data. Daily OHLCV for the current S&P 500 and Nasdaq 100 constituents (514 symbols), split/dividend adjusted (EODHD). Because the universe is today's membership, results carry survivorship bias — and it is stronger here than in the India study: constituents of major US indices are by construction the era's winners. Stocks delisted or dropped during the window are absent. A production engine would use effective-dated membership.

Execution. Signals are computed at each close; fills occur the next session at the OHLC/4 average price with 0.2% slippage each way. Position size is 4% of portfolio equity at entry, whole shares only, max 25 positions, no leverage. Idle cash earns nothing. Commissions are not modeled (near zero at US retail brokers); SEC/TAF fees and borrow costs are immaterial for a long-only book at this turnover.

Taxes. 25% flat on net realized gains each December 31 with loss carry-forward — a deliberate simplification mirroring the India report, not US tax advice. Actual short/long-term capital-gains treatment depends on holding period and bracket. All realized gains in the window are taxed; \$12.1k of losses remain as carry-forward at the window end.

Validity. The dip-entry cohort passed the house 6-check validity protocol on Indian data (NSE 500, 2012-2026). The protocol has NOT been run on the US panel; what is shown here is an out-of-sample portfolio-level test of the India-selected rules with one deviation — the trailing stop is dropped, per the US grid in RESULTS_US.md, which is itself a selection made on this same data (treat with corresponding caution).

Interpretation. This pre-AI-boom window spans four stress episodes — the Feb 2018 vol shock, the Q4 2018 selloff, the Mar 2020 COVID crash (the -37% max drawdown), and the 2022 bear (the one losing year, -12.7%, with losses carried forward). The edge is concentrated in 2017/2019/2020; 2021 was nearly flat as mega-cap leadership churned the momentum ranks. A single call — TSLA, entered Nov 2019 — returned +770% and drives a large share of the window's profit; the top 5 open positions (PWR, AZO, IRM, MPC, STLD) are 33% of final equity.

The strategy's concentration in whatever leads the tape is by design — momentum-rank slot priority — but a different regime would produce materially different results. Past performance, simulated or otherwise, does not guarantee future results.